

IVUS Calibration & Characterization Protocol

Volcano s5i / Eagle Eye Gold · bench protocol for E1–E9

IVUS Calibration & Characterization Protocol — Volcano s5i

This document is the **instrument-agnostic** bench protocol for the experiments needed to populate the simulator's per-instrument YAML config. The protocol itself is owned by this simulator repo; the per-instrument fitting code that implements it (and the resulting calibrated YAML files) lives next to this repo at `../instrument-calibration/<instrument-id>/` — see e.g. `../instrument-calibration/p035_visions/` for the Visions PV .035 / Volcano s5i calibration that is the running example throughout this document.

The protocol is organized as nine experiments, **E1–E9**. E1–E8 feed the per-frame raysim simulator; E9 feeds the deferred motion / acquisition layer. Each experiment includes:

1. Purpose — which CSV/YAML fields it produces.
2. Equipment list (with tolerance thresholds).
3. Setup diagram.
4. Console / device settings.
5. Step-by-step procedure.
6. Data to acquire.
7. Analysis — exact formulas / methods used to extract each parameter.
8. Acceptance criteria.

All experiments are run with the same Volcano s5i console, the same catheter, and at the **same water/phantom temperature (22 ± 1 °C)** unless otherwise noted. Speed of sound in degassed water at 22 °C is taken as **c_{water} = 1488 m/s** (Marczak 1997).

IVUS imaging geometry — read this before designing fixtures

These experiments are **specific to the Volcano s5i 64-element solid-state synthetic-aperture (SA) IVUS probe** (Eagle Eye Gold / Platinum family) and differ from external-probe ultrasound calibration in several important ways:

No mechanical rotation. The 64 transducer elements are arranged in a ring on the side of the catheter near the distal tip. The device generates a 360° image by SA-beamforming across these elements, *not* by rotating a single element. This means: - Per-azimuth PSF can vary because different sub-aperture combinations are used to reconstruct different angular bins. - "Rotating the catheter" only rotates the element ring as a rigid body; it does not change how the beamformer operates. It is still useful for cross-element averaging (the same target gets imaged by different element subsets) but is not a substitute for sampling multiple azimuths.

Beam is radial. Beams are emitted perpendicular to the catheter long axis, in a plane (the imaging plane) that is also perpendicular to the catheter axis. Targets must be placed at a known **radial distance** from the catheter; "depth" in the simulator and in this document means radial distance.

Two view conventions are used in the figures: - **Cross-section view** (looking down the catheter long axis): catheter appears as a small disc with a gold ring (the 64-element array). Targets are positioned in 2D with (radial distance, azimuth). Used by **E1, E2, E4, E5, E6, E8**. - **Side view** (catheter long axis horizontal in page): used by **E3** because the elevation direction is the catheter long axis.

Wires and rod targets must be parallel to the catheter long axis, so they appear as point-like scatterers in the imaging plane (cross-section view).

Lateral resolution scales with depth. Because the lateral axis of the PSF is an arc length at radius z , $\text{lateral_FWHM_mm}(z) = z \cdot \theta_FWHM(z)$. Even with a perfectly Gaussian beam, the millimetre-FWHM grows linearly with depth purely from geometry — analyses must convert between angular FWHM and arc-length FWHM consistently.

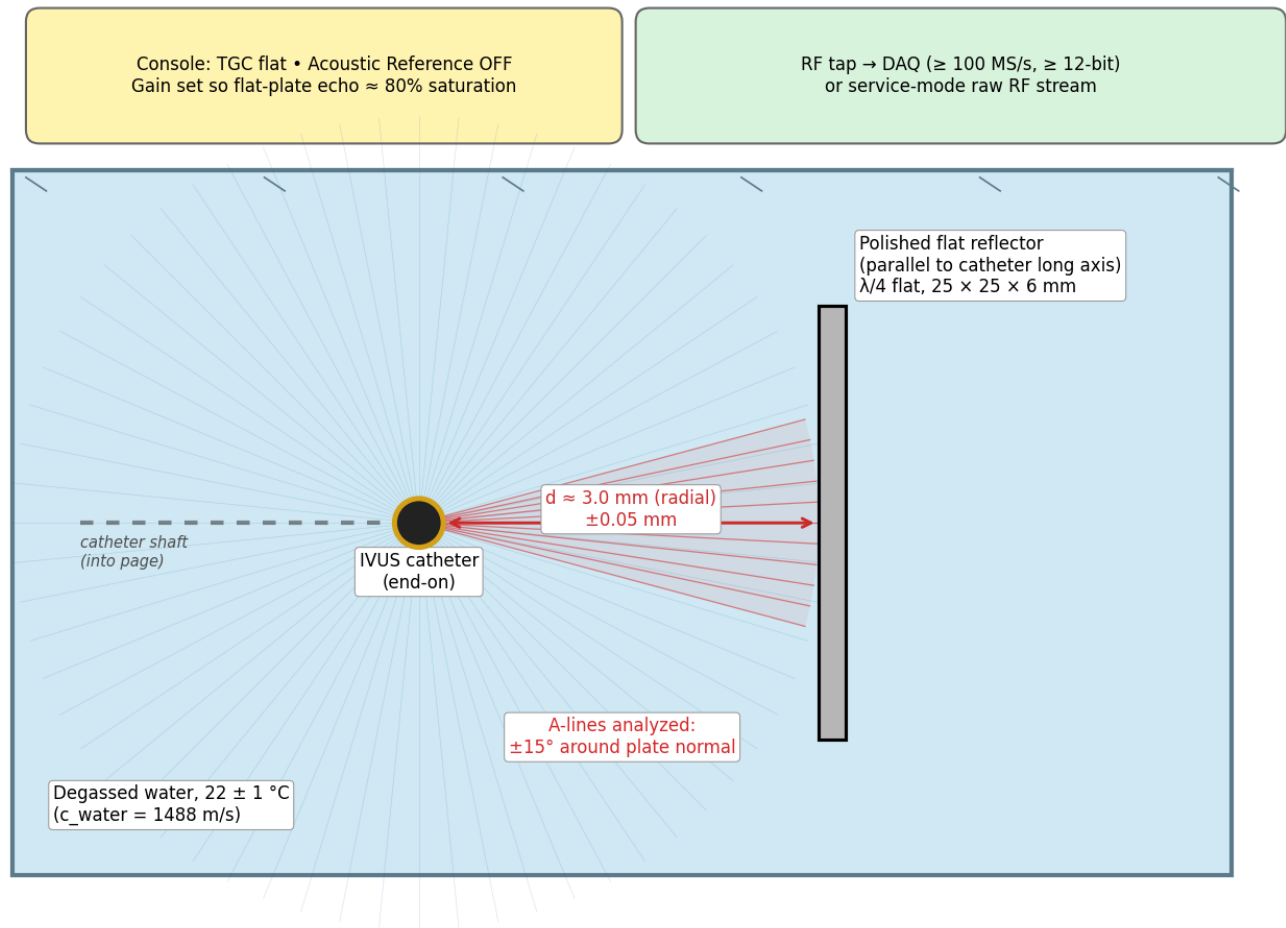
Elevation = along the catheter long axis. When measuring slice thickness, the target must move along the catheter shaft, not within the imaging plane.

Common equipment

Item	Spec	Tolerance / Notes
Volcano s5i console + Eagle Eye Gold (20 MHz) catheter	service-mode access	Same unit and same catheter S/N for all experiments
Water tank	$\geq 300 \times 200 \times 150$ mm, non-reflective lining	walls ≥ 30 mm from catheter tip in any direction
Degassed deionized water	dissolved $O_2 \leq 4$ ppm	let sit ≥ 24 h after degassing; $T = 22 \pm 1$ °C
3-axis micrometer / motorized stage	XYZ travel ≥ 25 mm	step resolution ≤ 25 μ m, repeatability ≤ 50 μ m
Catheter fixture	rigid clamp, axial alignment $\leq \pm 0.1$ mm	does not occlude the imaging plane
RF capture	DAQ ≥ 100 MS/s, ≥ 12 -bit (e.g. NI PXIe-5160)	direct RF tap on console back panel; if unavailable, use service-mode raw RF stream
Frame grabber	DVI/HDMI capture of console screen, $\geq 1080p$ @ 30 fps	for grayscale-only analyses (E7 fallback, E4 fallback)
Calibrated thermometer	± 0.2 °C	log T at the start of every session

E1 — Pulse-Echo Capture (flat reflector)

E1 — Pulse-Echo Capture (cross-section view; flat reflector parallel to catheter axis)



Produces: `probe.pulse_duration_cycles`, `probe.impulse_response_path`, `sim.buffer_size`, `sim.sampling_freq_mhz`, `validates` `probe.frequency_mhz`.

IVUS-specific note. Because the beam is radial, the reflector is a flat plate **parallel to the catheter long axis**, placed at a known radial distance from the catheter. Only A-lines whose azimuth is within $\sim \pm 15^\circ$ of the plate's surface normal are used in the analysis (those are the angles at which the reflection is near-normal-incidence and specular). All other A-lines see free water within `t_far` and are ignored.

Equipment

Item	Spec	Tolerance
Polished metal reflector (304 stainless or 6061 Al)	25 × 25 × 6 mm thick block, mounted parallel to the catheter long axis	flatness $\leq \lambda/4 \approx 19 \mu\text{m}$ (20 MHz in water); surface roughness $R_a \leq 0.4 \mu\text{m}$
Reflector tilt mount	goniometric, 2 axes; one axis is rotation about the catheter long axis (sets which catheter azimuth sees the plate normally), the other is in-plane tilt to sweep the normal	resolution $\leq 0.1^\circ$, range $\pm 10^\circ$
Reflector standoff	known radial distance <code>d</code> from catheter element ring	$d = 3.0 \text{ mm} \pm 0.05 \text{ mm}$
Catheter mount	rigid clamp, holds catheter horizontal so its long axis is fixed and the element ring is at a known location	axial position $\leq \pm 0.1 \text{ mm}$; allows the catheter to be rotated by 0/90/180/270° about its own long axis between captures

Console settings

Setting	Value
Imaging mode	IVUS B-mode
Catheter	Eagle Eye Gold (20 MHz)
Diameter	8 mm (smallest available; minimizes downsampling)
Gain (slider)	adjusted so the flat-reflector echo peaks at ~80% of full scale (no clipping)
TGC sliders	all centered (flat / disabled)
Acoustic Reference	OFF
Speckle reduction / smoothing	OFF
Compounding	OFF

Procedure

1. Mount the catheter horizontally in the fixture so the element ring is centered in the tank. Insert tip 50 mm into the water (≥ 30 mm clearance to walls).
2. Mount the reflector on the goniometric stage with its surface plane parallel to the catheter long axis and its surface normal pointing radially toward the catheter at azimuth $\theta_0 = 0^\circ$. Set the radial distance $d = 3.0$ mm from the element ring (use a calibrated depth gauge).
3. **Tune tilt**: rotate the reflector $\pm 2^\circ$ in each axis (in-plane tilt and tilt-about-catheter-axis) and record the peak A-line amplitude at the angular bin closest to θ_0 . Set the tilt to the maximum-amplitude position (normal incidence). Acceptance: peak amplitude must be within 1 dB of the maximum found during the sweep.
4. Verify d via time-of-flight: $d_{\text{est}} = c_{\text{water}} \cdot t_{\text{peak}} / 2$. Adjust the stage if $|d_{\text{est}} - 3.0| > 0.05$ mm.
5. Capture **300 frames** at static position. Within each frame, only the A-lines at azimuths $\theta \in [\theta_0 - 15^\circ, \theta_0 + 15^\circ]$ are kept; the rest see free water and are discarded. Average across the kept A-lines and across the 300 frames to obtain the impulse response.
6. Repeat steps 3–5 at $d = 5.0$ mm and $d = 8.0$ mm.
7. **Element-coverage replicate** (specific to the SA probe): rotate the catheter about its long axis by 90° , 180° , and 270° and repeat steps 3–5 at $d = 3.0$ mm. This re-images the same plate using different SA element subsets. Compare the four impulse responses; if they differ in pulse-duration or center-frequency by more than the acceptance criteria below, store all four (the simulator will use the mean and the device-side variability becomes an additional uncertainty input).

Data to acquire

- `e1_d3mm_rf.npy` — raw RF, shape (300, N_{samples})
- `e1_d5mm_rf.npy`, `e1_d8mm_rf.npy`
- `e1_metadata.json` — fs (claimed), gain, TGC, T_{water} , reflector tilt residual.

Analysis

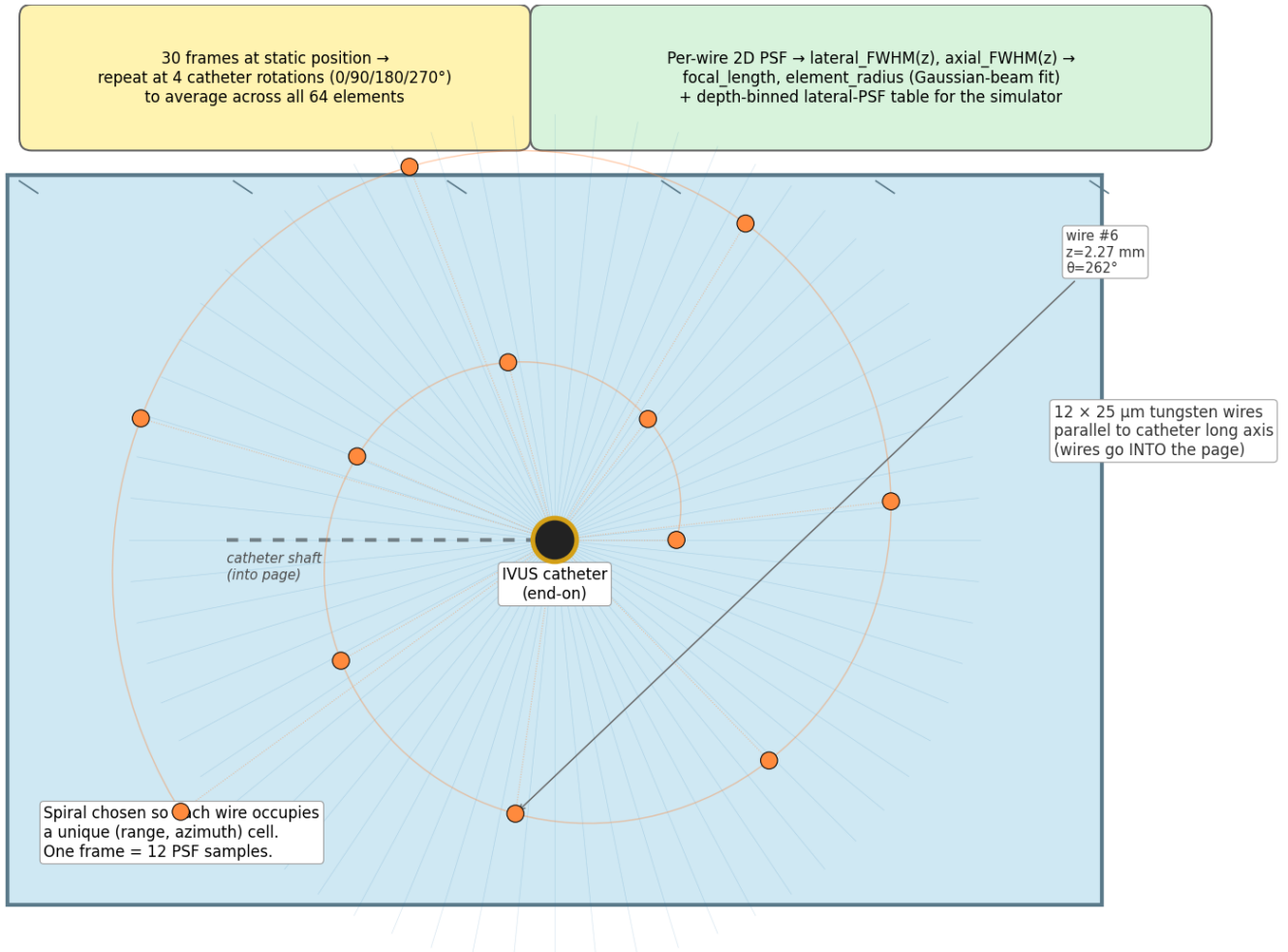
Output	Method
impulse_response	$h(t) = \text{mean}_n \text{ RF}(n, t)$ over 300 A-lines from the $d = 3$ mm capture, windowed $\pm 2 \mu\text{s}$ around the peak. Save as 1D float to <code>probe.impulse_response_path</code> .
frequency_mhz (verify)	$f_0 = \text{argmax} \text{FFT}(h) $. Must agree with manual within $\pm 5\%$.
pulse_duration_cycles	BW _{-6dB} from $ \text{FFT}(h) $, then $n = f_0 / \text{BW}_{-6\text{dB}}$. Cross-check by counting envelope zero-crossings within the -20 dB envelope.
sampling_freq_mhz	$f_s = N_{\text{samples_between_d5_and_d3}} \cdot c_{\text{water}} / (2 \cdot (d_5 - d_3))$. Should agree with the claimed f_s within $\pm 1\%$.
buffer_size	<code>N_samples</code> at the largest tested <code>t_far</code> for the configured Diameter.

Acceptance criteria

- Reflector tilt residual $\leq 0.5^\circ$ (peak amplitude within 1 dB of the maximum found during tilt sweep).
- d agreement (depth-gauge vs TOF): ≤ 0.05 mm.
- Coefficient of variation of A-line peak amplitude across 300 captures $\leq 2\%$.
- f_0 stable to within ± 0.2 MHz across the three depths.

E2 — Spiral Wire-Phantom 2D PSF

E2 — Spiral Wire-Phantom (cross-section view; samples PSF at multiple azimuths x radii)



Produces: `probe.element_radius_mm`, `probe.focal_length_mm`, depth-dependent lateral PSF table, F-number derivative, azimuthal-uniformity QC.

IVUS-specific note. A linear / radial wire arrangement (all wires on one radius) only samples a single azimuth and is sensitive to whichever SA-element subset reconstructs that bin. We instead use a **spiral arrangement** so a single capture samples PSF at multiple (range, azimuth) pairs. All wires are **parallel to the catheter long axis** (i.e. perpendicular to the imaging plane).

Equipment

[illegible]

Building the spiral fixture. A 3D-printable design is provided — see [Appendix A — Phantom & Fixture Construction](#). Briefly: print 2 × `hardware/wire_spiral_disc.stl` (80 mm OD × 5 mm) and 3 × `hardware/wire_spiral_standoff.stl` (or substitute commercial M3-F/M3-F threaded standoffs with M3 × 6 mm screws), thread the chosen filament through each pair of corresponding holes, tension by hand, and lock at each disc face — small bead of cyanoacrylate in the disc-top counterbore for any material; nylon can additionally be heat-melted into a ball against the disc face for a CA-free reversible lock.

Console settings

Identical to E1, with two changes: - Diameter set to the largest value that keeps every wire well-resolved at maximum z (e.g. 12 mm for a sweep up to $z = 6$ mm). - Gain set so the deepest wire echo is ≥ 30 dB above noise but the shallowest is $\leq 95\%$ saturated.

Procedure

1. Center the spiral fixture on the catheter; verify the catheter axis is on the spiral center to within ± 0.1 mm (visual or micrometer check).
2. Capture **30 frames** at static position. One frame contains a 2D PSF sample for each of the 12 wires.
3. **SA element-coverage replicate.** Rotate the catheter about its long axis by 90° , 180° , and 270° (the spiral fixture stays still). Capture 30 frames at each rotation. This re-images each wire with different SA element subsets.

Total: 12 wires × 4 rotations × 30 frames = 1440 PSF samples on a non-redundant (`range`, `azimuth`, `element-subset`) grid.

Data to acquire

- `e2_rotYYY_rf.npy` — RF, shape (30, N_{az} , N_{samples}), one file per catheter rotation.
- `e2_wire_positions.csv` — wire index, expected r (mm), expected θ (deg) for every wire.
- `e2_metadata.json` — gain, f_s , T_{water} , wire diameter, fixture S/N, tension log.

Analysis

For each (`rotation`, `wire`) pair:

1. Locate the wire echo: peak in the polar B-mode within a $\pm 5^\circ$ azimuthal window and ± 0.3 mm radial window of the expected (`r_n`, `theta_n`).
2. Extract a 3 mm (radial) × 30° (azimuthal) patch around the peak in **envelope** image space.
3. Average the patch across the 30 frames → 2D PSF for that (`rotation`, `wire`).
4. Convert the azimuthal axis to arc length: `arc_mm` = $r \cdot \theta_{\text{rad}}$ where r is the measured radial position of the wire.
5. Fit a 2D Gaussian (axial × arc-length) to extract `axial_FWHM(z)` and `lateral_FWHM_mm(z)` = `arc_FWHM(z)`.
6. Average the four rotations per wire to suppress per-element-subset variation; report the mean and σ .

Output	Method
<code>focal_length_mm</code>	$F = \text{argmin}_z \text{lateral_FWHM_mm}(z)$; parabolic fit on the three closest wires.
<code>element_radius_mm</code>	Fit Gaussian-beam waist $w(z) = w_0 \cdot \sqrt{1 + (z - F)^2 / z_R^2}$ to the rotation-averaged $w_{\text{meas}}(z) = \text{lateral_FWHM_mm}(z) / 1.177$. With $w_0 = \lambda \cdot F / (\pi \cdot a)$ and $z_R = \pi \cdot w_0^2 / \lambda$, solve for a (<code>element_radius_mm</code>). λ uses c_{water} at the measured tank temperature.
<code>f_number</code> (derived)	$F\# = F / (2a)$. Sanity check: should be in [1.5, 4].
<code>psf_lat_2d</code> lookup table	Save the depth-binned lateral PSF (rotation-mean) directly for use by <code>convolve_columns_depth_dependent</code> .
Azimuthal-uniformity QC	Per-wire σ across the four rotations vs the mean — reports SA per-element-subset variability.

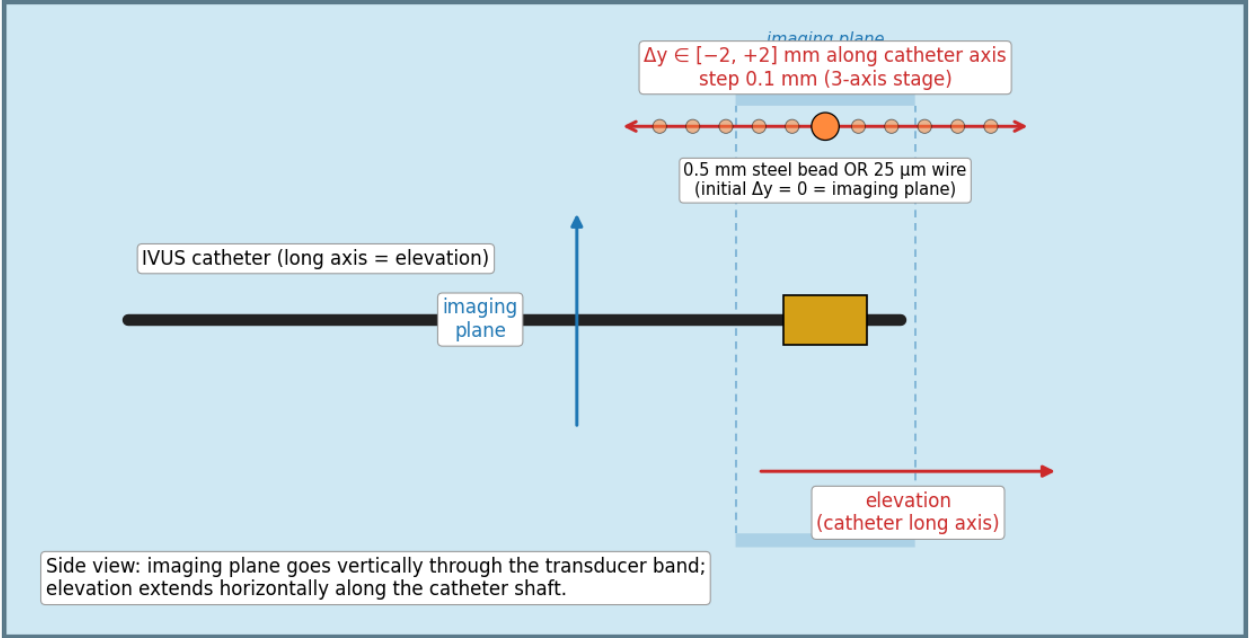
Acceptance criteria

- Per-wire echo SNR ≥ 30 dB.
- Spiral-center alignment error ≤ 0.1 mm (verified as: predicted vs measured r agreement across all 12 wires ≤ 0.1 mm RMSE).
- Per-rotation `lateral_FWHM(z)` consistency: σ across 4 rotations $\leq 10\%$ of the mean at every wire.
- Gaussian-beam fit residual $\leq 5\%$ across the 12 wires.

E3 — Slice-Thickness Sweep (elevational PSF)

E3 — Slice-Thickness (side view; elevation = catheter long axis)

Record peak A-line amplitude at the bead's azimuth vs $\Delta y \rightarrow$ elevational FWHM = elevational_height_mm



Produces: `probe.elevational_height_mm` and the elevational PSF profile.

IVUS-specific note. For an IVUS probe the "elevation" direction is **along the catheter long axis**, not perpendicular to it. The bead must be translated parallel to the catheter shaft, not within the imaging plane. The figure above shows the side view to make this explicit.

Equipment

Item	Spec	Tolerance
Target	25 μm tungsten wire OR 0.5 mm steel bead on monofilament	bead diameter $\leq \lambda$ at the focal depth
3-axis micrometer stage	as in "Common equipment"	Δy step 0.1 mm, range ± 2.0 mm — Δy is along the catheter long axis
Catheter mount	holds catheter long axis horizontal and stationary	axial play $\leq 25 \mu\text{m}$ during the sweep

Console settings

Same as E2.

Procedure

1. Mount the catheter horizontally with the element ring at a known position. Position the bead at radial distance equal to the focal length F (from E2), at azimuth 90° above the catheter axis (so its echo is in a clean A-line bin).
2. Align the bead's elevation (along-axis position) with the center of the imaging plane ($\Delta y = 0$): sweep the bead along the catheter long axis over ± 0.2 mm in 0.05 mm steps and pick the position of maximum A-line peak amplitude. This is $\Delta y = 0$.
3. Sweep $\Delta y \in [-2.0, +2.0]$ mm along the catheter long axis, 0.1 mm steps. At each Δy : capture **30 frames** and log the peak A-line amplitude at the bead's azimuthal bin.
4. Optionally repeat at $z = F/2$ and $z = 2F$ to characterize depth-dependence of slice thickness.

Data to acquire

- e3_dyXXX_rf.npy
- e3_amplitude_vs_dy.csv — Δy (mm), peak amplitude, std.

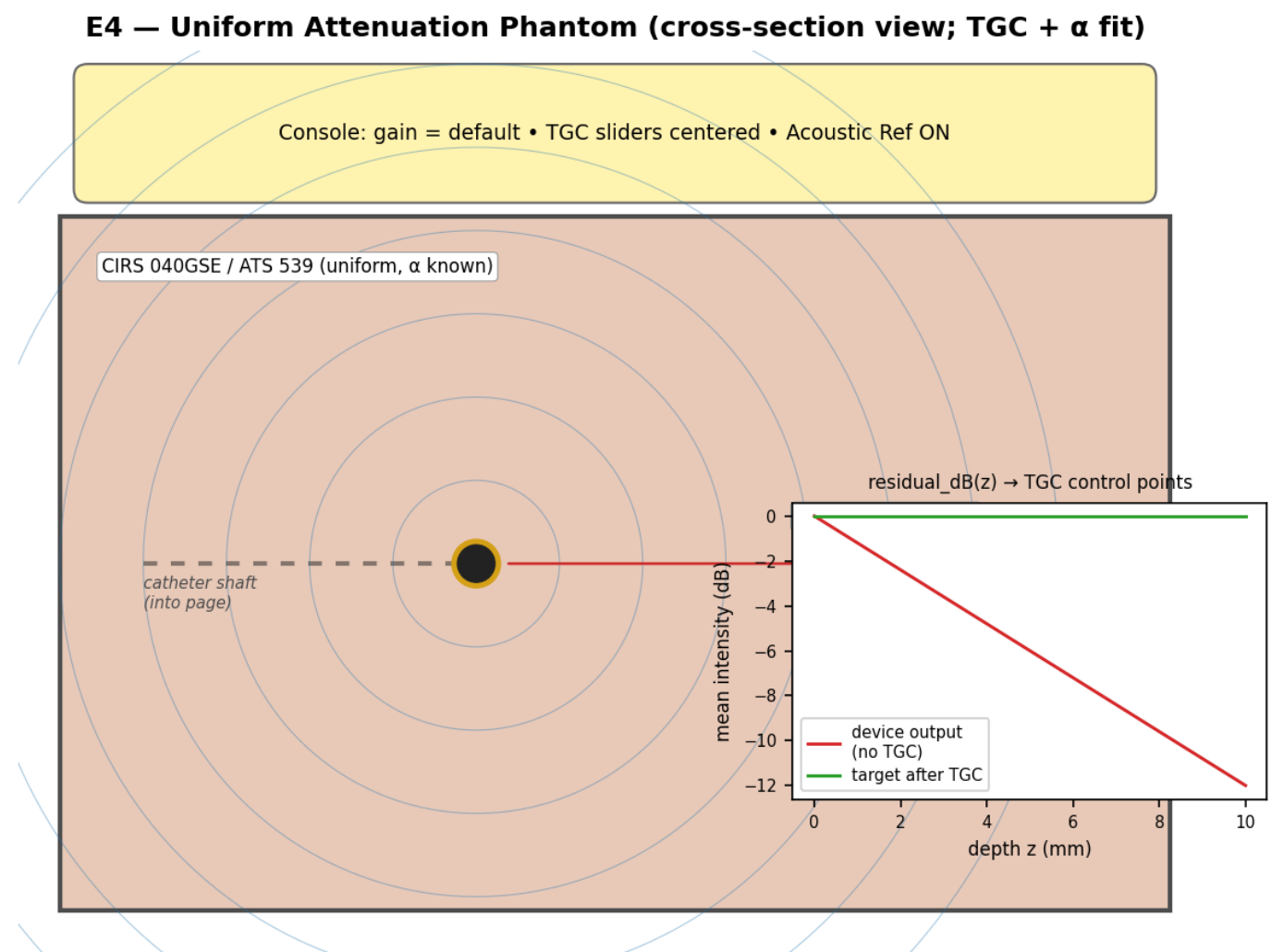
Analysis

Output	Method
elevational_height_mm	FWHM of peak_amplitude(Δy) curve.
Elevational PSF	Save peak_amplitude(Δy) normalized to 1.0 at $\Delta y = 0$ (used for sensitivity-weighting the elevation samples in the simulator).

Acceptance criteria

- Profile symmetry: $|FWHM_{left} - FWHM_{right}| / FWHM \leq 10\%$.
- Peak repeatability across 30 frames at $\Delta y = 0$: $CV \leq 2\%$.

E4 — Uniform Attenuation Phantom (TGC + α)



Produces: `processing.tgc_control_points`, `materials[].attenuation_db_per_cm_mhz` prior, refined `probe.speed_of_sound_mm_per_us`.

Equipment

Item	Spec	Tolerance
Uniform tissue-mimicking phantom	CIRS Model 040GSE, 0.5 dB/cm/MHz, with 2.7 F-compatible IVUS channel; or DIY agar/graphite phantom — see Appendix A.2	α calibrated to ± 0.05 dB/cm/MHz (manufacturer) or ± 0.10 dB/cm/MHz (DIY, by transmission-substitution); T-corrected
Phantom temperature	22 ± 1 °C	equilibrate ≥ 4 h

Console settings

Setting	Value
Diameter	16 mm (= manual default for service tests)
Gain	50 (manual default)
TGC	all sliders centered
Acoustic Reference	ON
Speckle reduction	OFF

Procedure

1. Insert the catheter into the phantom's IVUS channel; advance to the marked center.
2. Allow ≥ 60 s of equilibration after movement.
3. Capture **30 frames** at static position with the settings above.
4. Repeat at 3 axially separated channel positions (5 mm apart) — each is a replicate.

Data to acquire

- `e4_posK_rf.npy` for $K = 1..3$ — RF or polar B-mode.
- `e4_metadata.json` — phantom S/N, α_{spec} , T_{phantom} , gain, TGC, console build.

Analysis

1. Convert each frame to polar B-mode (angle $\times$ depth).
2. Mask out the inner 0–2 mm (ring-down zone) and the outer 0.5 mm (boundary echoes).
3. Compute mean intensity vs depth $I(z) = \text{mean}_{\theta} B(\theta, z)$.
4. Convert to dB and average across the 3 replicate positions.

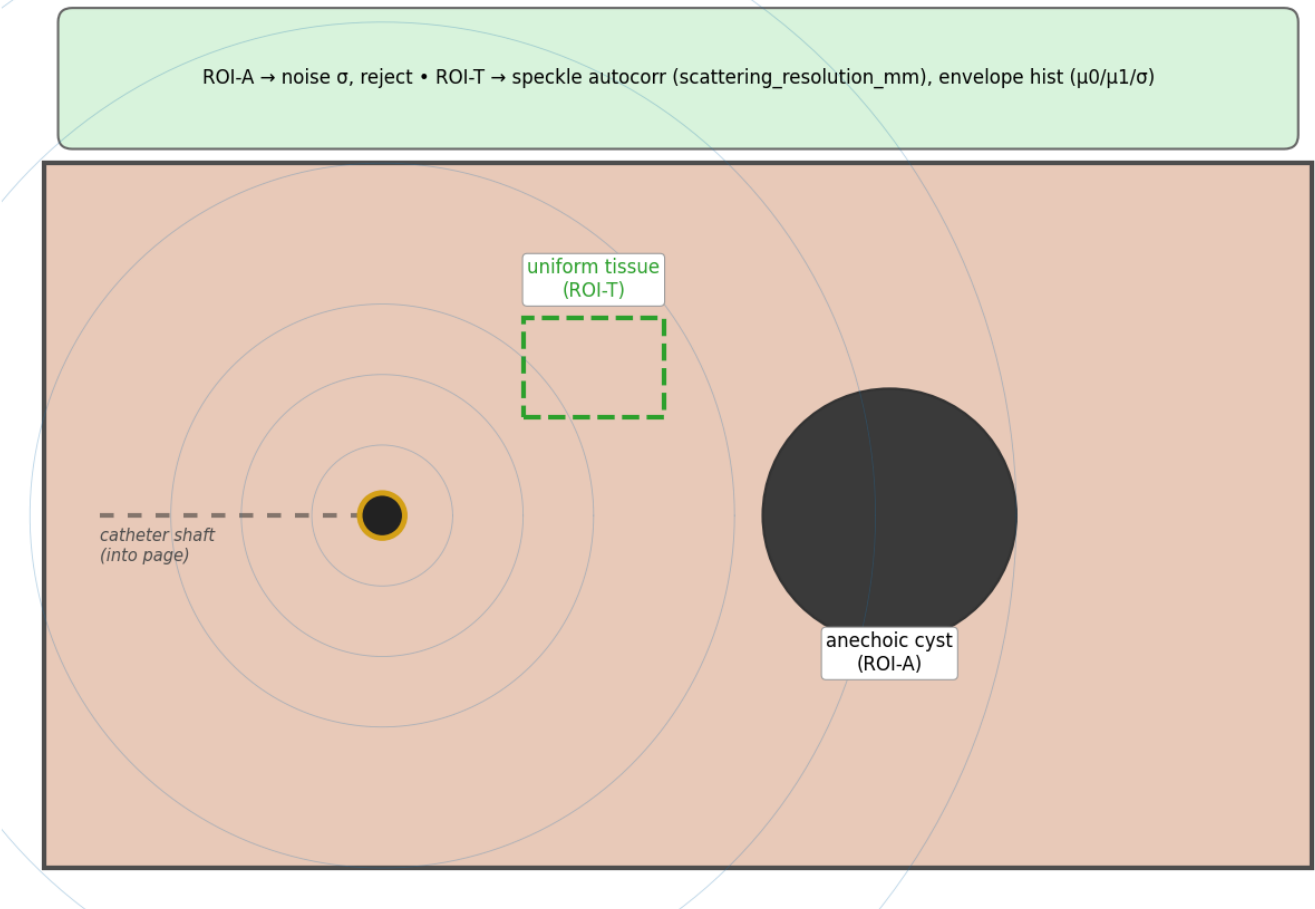
Output	Method
Effective α (verify phantom)	Slope of $I_{\text{dB}}(z) \div (2 \cdot f_0)$ gives effective α (dB/cm/MHz). Must agree with manufacturer α to within ± 0.10 dB/cm/MHz.
<code>tgc_control_points</code>	The TGC needed to flatten $I_{\text{dB}}(z)$ is $g(z) = -I_{\text{dB}}(z) + I_{\text{dB}}(0)$. Sample at depths {0.0, 1.5, 3.0, 4.5, 6.0, 8.0} mm to define 6-point control polyline. Convert depths to cm.
<code>speed_of_sound_mm_per_us</code>	If the phantom contains a known-distance feature (CIRS 040GSE has line-target wire spacing), measure echo TOF and back out c . Use as a refinement to the 1.54 mm/ μ s default.

Acceptance criteria

- Replicate-to-replicate variation in $I_{\text{dB}}(z) \leq 1$ dB at every depth.
- α agreement with phantom spec: within ± 0.10 dB/cm/MHz.

E5 — Cyst Phantom (speckle, noise, reject)

E5 — Cyst Phantom (cross-section view; speckle, noise, reject)



Produces: `processing.noise.type`, `processing.noise.sigma`, `processing.scattering_resolution_mm`, `processing.scatter_integral_scale`, `processing.reject_db`, prior on per-tissue scattering parameters (`mu0`, `mu1`, `sigma`).

Equipment

Item	Spec	Tolerance
Cyst-bearing tissue-mimicking phantom	CIRS 040GSE (anechoic cylinders 2, 4, 6 mm dia.) or equivalent; or DIY phantom with PTFE-rod cyst voids — see Appendix A.3	manufacturer-calibrated; or DIY built using the same recipe as A.2

Console settings

Identical to E4 (gain = 50, TGC centered, AR ON), repeated at gain = 20 and gain = 68 for noise-floor / saturation characterization.

Procedure

1. Position the catheter so a 4 mm anechoic cyst is in the field at $z \approx 5$ mm.
2. Capture **30 frames** at gain = 50.

3. Repeat at gain = 20 and gain = 68 (same catheter position).

Data to acquire

- `e5_gainXX_rf.npy` for $XX \in \{20, 50, 68\}$.
- ROI definitions: `roi_A` (anechoic interior, $\geq 3 \times 3 \text{ mm}^2$ of cyst lumen, no boundary), `roi_T` (uniform tissue, $\geq 5 \times 5 \text{ mm}^2$, away from any cyst or boundary).

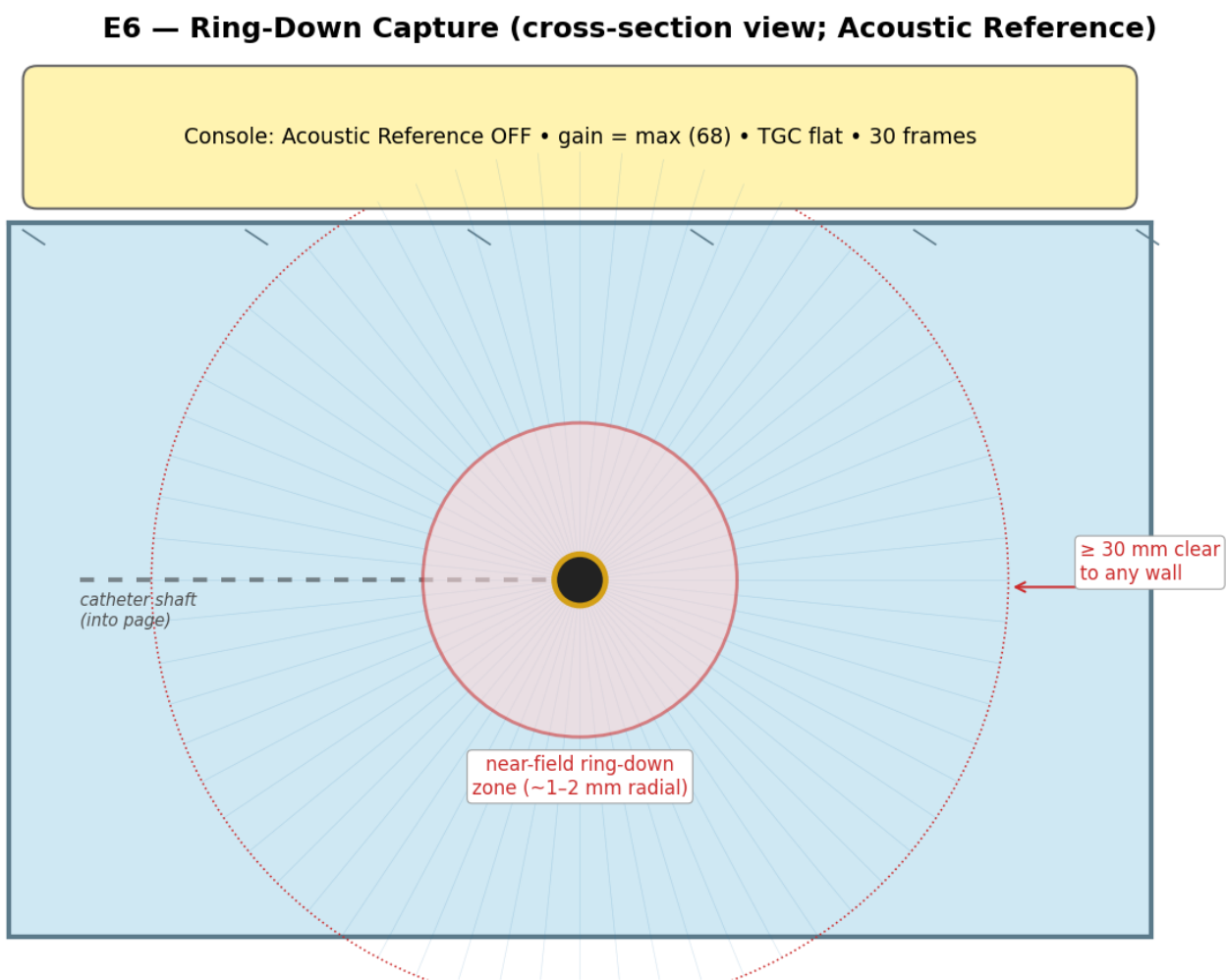
Analysis

Output	Method
<code>noise.sigma</code> (linear units)	std of pre-envelope samples in <code>roi_A</code> at gain = 20 (clean noise floor). For envelope-only data, use Rayleigh-fit σ .
<code>noise.type</code>	KS-test of <code>roi_A</code> envelope histogram vs Rayleigh / Gaussian / Nakagami. Pick lowest-D test.
<code>reject_db</code>	<code>dB_below_T_at_roi_A_50_pct</code> — i.e. the dB level at which 50% of <code>roi_A</code> pixels fall below display threshold.
<code>scattering_resolution_mm</code>	2D autocorrelation of <code>roi_T</code> ; report the 1/e radius along radial and lateral axes (use mean for the scalar field).
<code>scatter_integral_scale</code> (provisional)	Run sim with measured material defaults; tune until simulated <code>roi_T</code> mean matches measured <code>roi_T</code> mean within 1 dB.
<code>mu0 / mu1 / sigma</code> priors	Fit a homodyned-K distribution to the <code>roi_T</code> envelope histogram. Map shape parameter $k \rightarrow \mu_0/\mu_1$ (literature mapping); RMS amplitude $\rightarrow \sigma$.

Acceptance criteria

- `roi_A` mean intensity drops by $\geq 30 \text{ dB}$ vs `roi_T` (confirms cyst is anechoic).
- KS p-value for chosen noise distribution ≥ 0.05 .
- Speckle autocorrelation length consistent across gain settings ($\text{CV} \leq 15\%$).

E6 — Ring-Down Capture (Acoustic Reference)



Produces: `processing.ring_down.amplitude`, `processing.ring_down.extent_mm`, `processing.ring_down.decay`, `processing.ring_down.waveform_path`.

Equipment

Item	Spec	Tolerance
Water tank	as in “Common equipment”	walls ≥ 30 mm clear of catheter in any direction
Catheter mount	suspends catheter centered in tank	catheter tip ≥ 30 mm from any wall, surface, or other object

Console settings

Setting	Value
Diameter	16 mm (full t_far range)
Gain	68 (max)
TGC	all sliders centered
Acoustic Reference	OFF (this is the key requirement)
Speckle reduction	OFF

Procedure

1. Suspend the catheter in the center of the tank, away from all walls and surfaces (≥ 30 mm clearance).
2. Wait ≥ 60 s for any tank vibrations / bubbles to settle.
3. Capture **300 A-lines $\times$ 30 frames** at the same angular position.
4. Save the device's on-board "Acoustic Reference" file via the service interface (this is the device's stored ring-down template — useful for cross-validation).
5. Repeat with Acoustic Reference **ON**, same conditions, 30 frames (delta capture for validation).

Data to acquire

- `e6_ar_off_rf.npy` — shape (30, N_{az} , N_{samples}).
- `e6_ar_on_rf.npy` — same, with AR ON.
- `e6_device_acoustic_reference.bin` — exported from console (if accessible).

Analysis

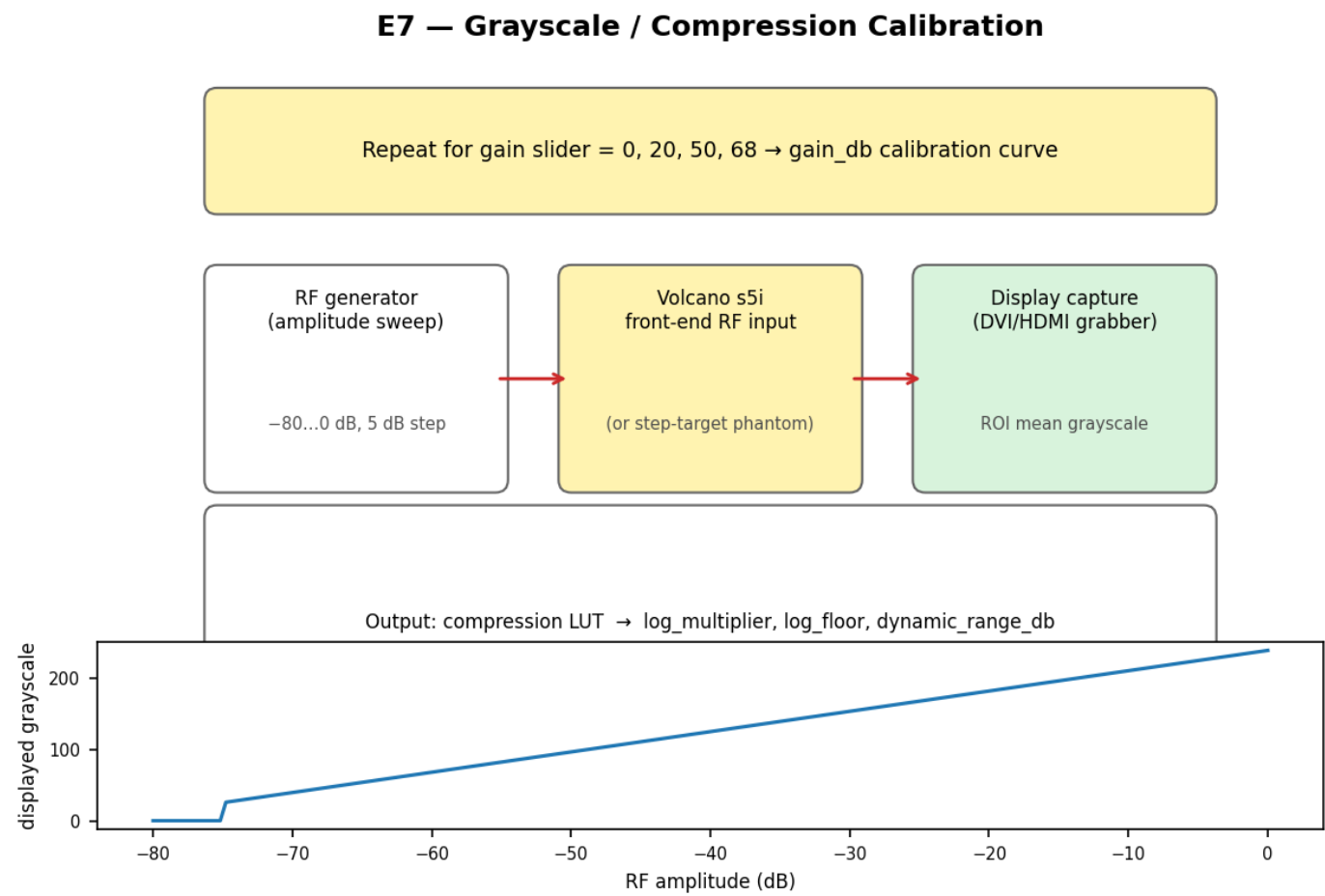
1. Compute `mean_aline = mean over angles and frames of |envelope(rf_off)|`.
2. Compute the noise floor σ_n from the deep portion ($z > t_{\text{far}} - 0.5$ mm).

Output	Method
<code>ring_down.amplitude</code>	<code>max_z mean_aline(z)</code> (typically near $z = 0$). Report in linear and dB-of-saturation.
<code>ring_down.extent_mm</code>	Smallest z at which <code>mean_aline(z) < 3 σ_n</code> .
<code>ring_down.decay</code>	Fit <code>A · exp(-z/τ)</code> and <code>A · 0.5(1 + cos($\pi z/L$))</code> to <code>mean_aline(z)</code> over <code>[0, extent]</code> . Pick lower-RMSE form. Save τ or L .
<code>ring_down.waveform_path</code>	Save <code>mean_aline(z)</code> as a 1D float on the simulator's 40 MHz grid (resample if device f_s differs).
Validation	<code>mean over angles and frames of envelope(rf_on)</code> should be within $\pm 10\%$ of <code>mean_aline(z) - device_acoustic_reference(z)</code> over <code>[0, extent]</code> .

Acceptance criteria

- Per-frame A-line variation in the inner 2 mm $\leq 5\%$ (the bias is deterministic).
- Validation step matches AR-ON capture within $\pm 10\%$.

E7 — Grayscale / Compression Calibration



Produces: `processing.compression_lut`, `processing.log_multiplier`, `processing.log_floor`, `processing.dynamic_range_db`, `processing.gain_db` (slider→dB mapping).

Equipment (preferred — RF injection)

Item	Spec	Tolerance
Programmable RF generator	DDS, ≥ 50 MHz BW, ≥ 14-bit amplitude	amplitude accuracy ≤ ±0.5 dB
Variable RF attenuator	0–80 dB, 1 dB steps	calibrated, ≤ ±0.2 dB step error
Front-end coupling jig	injects RF at the catheter connector in place of the active element	shielded, return loss ≥ 30 dB
Frame grabber	DVI/HDMI capture, lossless 8-bit	full 0–255 range preserved

Equipment (fallback — step phantom)

Item	Spec	Tolerance
Step-target tissue-mimicking phantom	CIRS 044, multi-step grayscale targets at 0/3/6/9/12/15/18 dB below background; or DIY 6-chamber concentric phantom — see Appendix A.4	manufacturer ±0.5 dB; DIY ±1 dB if α calibrated by transmission-substitution per chamber

Console settings

Setting	Value
Acoustic Reference	ON (default)
TGC	all sliders centered
Dynamic range / persistence	factory default
Speckle reduction	OFF

Procedure (RF injection)

1. Disconnect the catheter element; connect the RF generator through the coupling jig.
2. Inject a damped sinusoid at f_0 with amplitude A set by the attenuator.
3. For each gain setting $g \in \{0, 20, 50, 68\}$:

For each amplitude $A \in \{-80, -75, \dots, 0\}$ dB:

- Capture display ROI mean grayscale (200 × 200 px ROI at the depth of the simulated echo).
 - Wait ≥ 1 s for any temporal averaging on the device to settle.
5. Total: 4 gains × 17 amplitudes = 68 calibration points.

Procedure (fallback)

1. Place the step-phantom in the field; image so all step targets fit in one frame.
2. For each gain $g \in \{0, 20, 50, 68\}$: capture 30 frames, average, extract per-step mean grayscale and per-step nominal dB.
3. Use only the linear (non-saturated, non-clipped) portion of the curve.

Data to acquire

- `e7_gainXX_lut.csv` — columns: amplitude_dB, mean_grayscale, std_grayscale.

Analysis

Output	Method
<code>compression_lut</code>	At gain = 50, the mapping RF_amplitude (linear) → grayscale (uint8) is the stored LUT (256 entries). Linearly interpolate between the 17 calibration points. Save as JSON array.
<code>log_multiplier</code>	Slope of grayscale vs $20 \cdot \log_{10}(\text{amplitude_linear})$ in the linear region (typically $100 \leq \text{grayscale} \leq 200$).
<code>log_floor</code>	Largest amplitude at which grayscale = 11 (palette black floor; manual p.141).
<code>dynamic_range_db</code>	$20 \cdot \log_{10}(A_{\text{at_grayscale}=239} / A_{\text{at_grayscale}=11})$.
<code>gain_db(slider)</code>	For each gain g , find the dB shift in the calibration curve relative to $g = 0$. Output a 4-point table slider → dB.

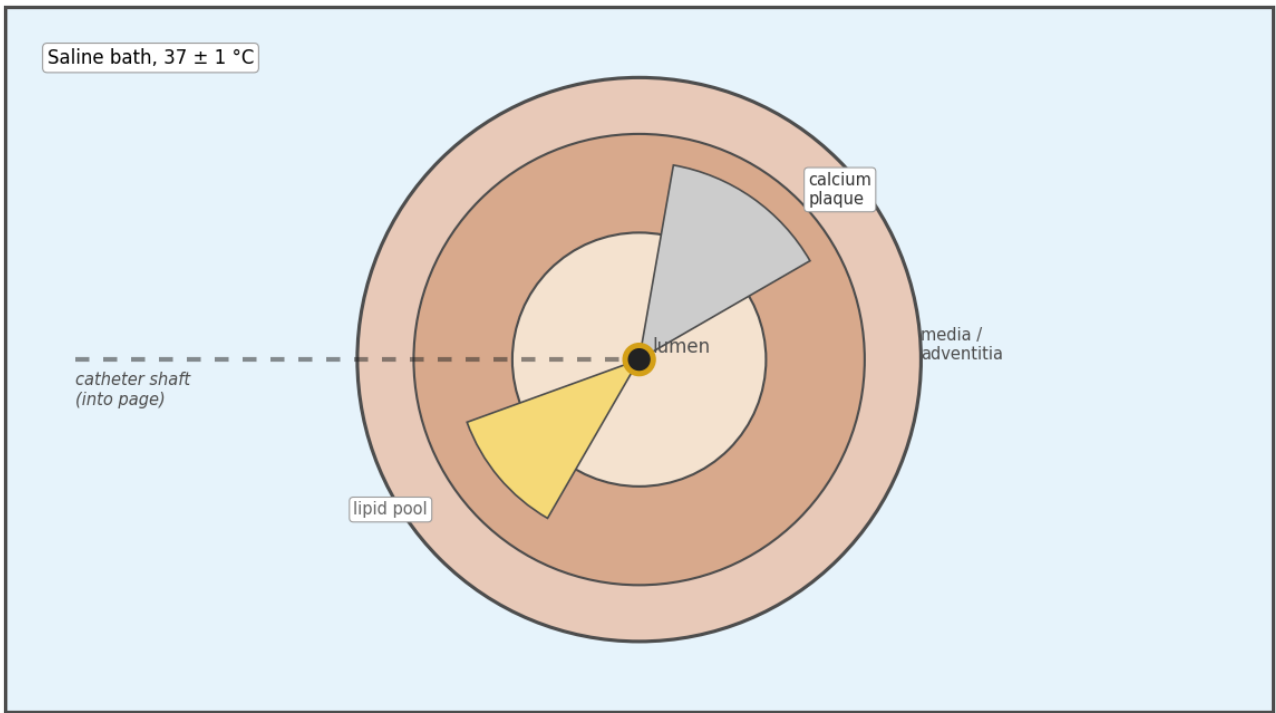
Acceptance criteria

- Calibration repeatability (ROI std at each amplitude) ≤ 1 grayscale unit.
- Monotonicity: LUT must be non-decreasing.
- gain_db curve linearity across the four sliders: residual ≤ 1 dB.

E8 — Tissue / Material Fit

E8 — Tissue / Material Fit (cross-section view; ex vivo or layered phantom)

Co-register IVUS RF with μ CT / histology $\rightarrow$ per-material $Z, c, \alpha, \mu_0/\mu_1/\sigma$, specularity



Produces: per-material `impedance_mrayl`, `speed_of_sound_m_per_s`, `attenuation_db_per_cm_mhz`, `mu0`, `mu1`, `sigma`, `specularity` for `lumen`, `vessel_wall` (intima / media / adventitia), `extravascular`, `calcium`, `fibrous`, `lipid_pool`.

Equipment

Item	Spec	Tolerance
Ex vivo arterial samples	porcine LAD or human cadaver coronary; or CIRS 067 IVUS plaque phantom; or DIY layered tissue surrogate — see Appendix A.5	preserved at 4 °C, used within 48 h of harvest; warmed to 37 °C for imaging
Saline bath	0.9% physiological saline	$T = 37 \pm 1$ °C; degassed
Sample fixture	IVUS-channel through a saline-filled trough; gentle vessel mounts	maintains lumen geometry
Reference modality	μ CT (≤ 50 μ m voxel) or histology (≤ 5 μ m/pixel)	co-registered to IVUS using fiducials (e.g. metal pins inserted before μ CT)

Console settings

Identical to E4 (gain = 50, TGC centered, AR ON).

Procedure

1. Mount the sample in the saline trough; insert the catheter into the lumen.
2. Acquire **30 frames** at each of 3 longitudinal positions per sample.
3. After IVUS imaging (without disturbing the sample) acquire μ CT or histology.
4. Co-register IVUS frames to μ CT/histology slices using the fiducial pins.
5. For each material class, manually segment ROIs $\geq 3 \times 3 \text{ mm}^2$ (or the largest possible).

Data to acquire

- `e8_sampleK_posJ_rf.npy`, `e8_sampleK_posJ_bmode.png`.
- `e8_sampleK_uct.nrrd` or histology TIFFs.
- `e8_sampleK_segmentation.nii.gz` — per-material masks co-registered to IVUS.

Analysis

For each material:

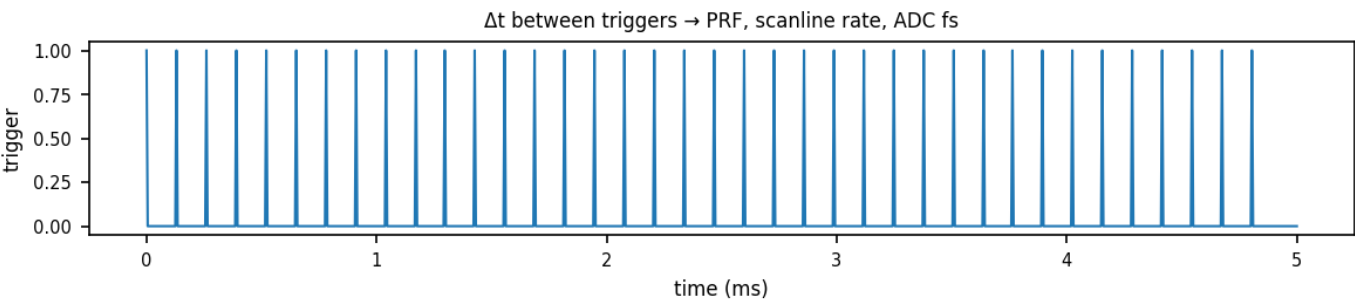
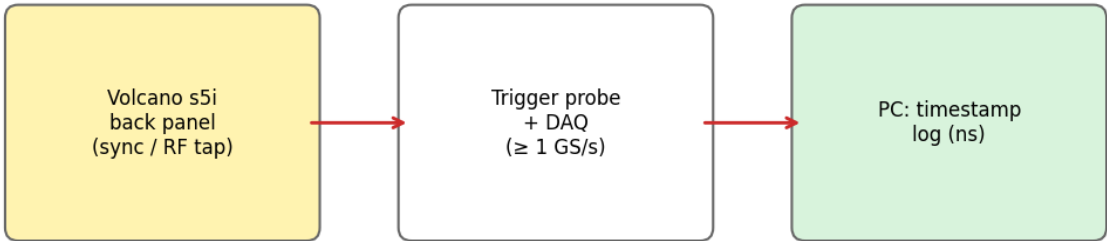
Output	Method
<code>impedance_mrayl</code>	Identify a clean interface between this material and a reference material (saline / known phantom). Measure reflection amplitude <code>R_meas</code> at normal incidence. Solve $Z = Z_{\text{ref}} \cdot (1 + R_{\text{meas}}) / (1 - R_{\text{meas}})$ (sign from phase). Use ≥ 5 interface segments per material; report median $\pm$ IQR.
<code>speed_of_sound_m_per_s</code>	TOF through known thickness (from μ CT). $c = 2 \cdot d_{\text{uct}} / \Delta t_{\text{echo}}$. ≥ 5 thickness measurements per material.
<code>attenuation_db_per_cm_mhz</code>	Within a homogeneous ROI $\geq 3 \text{ mm}$ thick, fit $\ln(\text{env}(z))$ vs depth. Slope / $(2 \cdot f_0) = \alpha$.
<code>mu0 / mu1 / sigma</code>	Homodyned-K fit to the envelope histogram of the material ROI (as in E5).
<code>specularity</code>	For the vessel wall: extract $R(\theta)$ at oblique segments where θ_{inc} can be computed from local wall normal (from μ CT). Fit $R(\theta) = R_0 \cdot \cos^n(\theta)$; <code>specularity</code> = n / N (mapped to the simulator's 0–1 range).

Acceptance criteria

- ≥ 3 independent samples per material class.
- Per-material parameter CV across samples $\leq 30\%$.
- μ CT–IVUS co-registration error $\leq 0.2 \text{ mm}$ (RMSE on fiducials).

E9 — Acquisition Timing Log (deferred layer)

E9 — Acquisition Timing Log (deferred layer)



Produces: PRF, frame_rate, scanline_rate, ADC sampling rate. These are *not* part of the per-frame raysim config but feed the motion / acquisition layer (see CSV "Deferred" section).

Equipment

Item	Spec	Tolerance
Trigger probe / sync tap	high-impedance probe on console's RF or sync line	bandwidth $\geq 2 \times f_0$
DAQ / oscilloscope	≥ 1 GS/s, ≥ 1 GHz analog BW, deep memory	timestamp jitter ≤ 1 ns

Console settings

Whatever clinical preset is being characterized (e.g. 30 fps, Diameter = 16 mm).

Procedure

1. Tap the sync line; capture 10 s of trigger pulses at the DAQ.
2. Repeat for each frame-rate preset (10, 15, 30 fps) and each Diameter setting of interest.

Data to acquire

- e9_presetXX_triggers.npy — float64 timestamps in seconds.

Analysis

Output	Method
PRF	Median of $1 / \Delta t_{\text{trigger}}$.
frame_rate	$1 / \Delta t_{\text{first_trigger_per_frame}}$ (frame boundaries detected by long Δt).
scanline_rate	$\text{frame_rate} \times \text{num_scanlines}$.
ADC sampling rate (cross-check)	If RF stream is also captured, count samples per known round-trip from E1 reflector.

Acceptance criteria

- PRF jitter (std/mean) $\leq 0.1\%$.
- Frame-rate measurement matches console label within ± 0.2 fps.

Summary table — what each experiment produces

Experiment	Days	Primary outputs
E1 — Pulse-echo (radial flat plate)	0.5	impulse response, pulse duration, fs, buffer size
E2 — Spiral wire-phantom PSF	1.0	element_radius, focal_length, depth-dependent lateral PSF, azimuthal-uniformity QC
E3 — Slice thickness (along catheter axis)	0.5	elevational_height
E4 — Attenuation phantom	0.5	TGC control points, α prior, refined c
E5 — Cyst phantom	0.5	noise σ , scattering_resolution_mm, scatter scale, reject, μ priors
E6 — Ring-down	0.5	ring_down amplitude / extent / decay / waveform
E7 — Grayscale / compression	0.5 (RF) / 1 (phantom)	compression LUT, log_multiplier, log_floor, dynamic range, gain map
E8 — Tissue fit	3+ (per sample lot)	per-material Z, c, α , $\mu_0/\mu_1/\sigma$, speculariry
E9 — Timing log	0.25	PRF, frame rate, scanline rate, ADC fs

Recommended order: E1 → E2 → E3 → E6 → E4 → E5 → E7 → E8, with E9 anytime.

Reproducibility

All schematic figures in this document are generated by:

```
PYTHONPATH=/path/to/matplotlib MPLBACKEND=Agg python3 tools/gen_calibration_figures.py
```

The hardware STL files (Appendix A) are generated by:

```
PYTHONPATH=/path/to/trimesh:/path/to/shapely:/path/to/manifold3d \  
python3 tools/gen_phantom_stl.py
```

The hardware preview renders are generated by:

```
PYTHONPATH=/path/to/trimesh:/path/to/matplotlib MPLBACKEND=Agg \  
python3 tools/gen_hardware_previews.py
```

All three scripts read the same shared dimensions, so editing one source of truth ([tools/gen_phantom_stl.py](#)) regenerates consistent geometry across STLs, renders, and protocol figures.

Appendix A — Phantom & Fixture Construction

This appendix describes how to build the phantoms called for in the experiments above. Wherever a commercial calibrated phantom (CIRS / ATS / Sun Nuclear) is available you should prefer it because the manufacturer provides traceable α and grayscale specifications. The DIY recipes below are validated reproductions of the literature standards (Madsen 1978, Madsen 1998, Ramnarine 2001, Browne 2003) and are sufficient for **simulator-input characterization** when paired with a one-time transmission-substitution α measurement (see A.6).

All printed parts are designed for FDM with a **0.4 mm nozzle**, **0.2 mm layer height**, **$\geq 30\%$ infill**, **≥ 3 perimeters**, and **PETG** or **ABS** filament (PLA is acoustically acceptable but warps in warm gel — avoid it for the molds). Acetone-vapor smoothing of ABS is recommended for mold surfaces if you want a glassy finish; otherwise expect mild ridging on the gel surface (acoustically harmless because it is below the resolution limit).

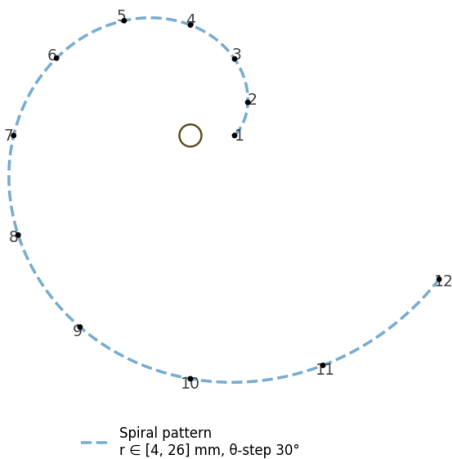
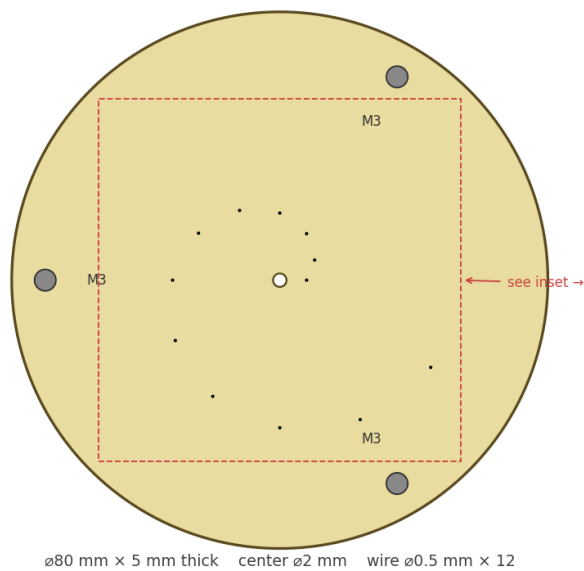
The Python source for every printed part is [tools/gen_phantom_stl.py](#) — edit dimensions there and rerun to regenerate the STLs.

A.1 — Wire-spiral fixture (E2)

Wire-spiral disc — 12 holes, 1-turn spiral, $r \in [4, 26]$ mm

Top view (printed disc, full scale)

Spiral pattern detail (zoomed)



STL files: - [hardware/wire_spiral_disc.stl](#) — print **2 ×**. - [hardware/wire_spiral_standoff.stl](#) — print **3 ×** (or substitute commercial M3 standoffs, see BoM option B). - [hardware/wire_spiral_assembly.stl](#) — visual reference only (do not print).

Disc geometry (matches the protocol's spiral parameters):

Feature	Value
Outer diameter	80.0 mm
Thickness	5.0 mm
Center catheter hole	Ø2.0 mm (øp-06 for the Ø3.0 mm Visiovis PV-035 catheter — measure your actual catheter ID with calipers and round out to 0.1 if necessary)
Wire holes	Ø0.5 mm × 12 through-holes, with a Ø1.0 mm × 0.4 mm CA-epi counterbore on the top face for the (symmetrical) lock-head. Listed: (0 mm, 0°) holes (0.0), (6.35), (8.89), (10.95), (12.125), (14.125), (16.18), (18.215), (20.245), (22.275), (24.305), (26.335)
Standoff holes	Ø3.2 mm × 3, on a Ø70 mm BCD at 90°/180°/270° (clearance for either the printable shouldered standoff's 3.1 mm top pin or an M3 screw thread)

Bill of materials:

Item	Spec	Qty
Printed disc (wire_spiral_disc.stl)	PETG, $\geq 30\%$ infill, ≥ 3 perimeters	2
Standoffs (option A — printable) wire_spiral_standoff.stl	PETG, $\geq 40\%$ infill, ≥ 4 perimeters; printed vertically (long axis along Z) for concentric pin tips	3
Standoffs (option B — commercial) M3-F/M3-F threaded standoff	stainless steel, hex flat-to-flat ≤ 5 mm, length 50 mm	3
M3 $\times$ 6 mm pan-head screws (option B only)	stainless	6
Wire — pick one material from this menu (all 12 wires same material)	See "Wire material trade-offs" below	12 $\times$ ≥ 250 mm
Cyanoacrylate adhesive	thin-CA (e.g. Loctite 416) — locks any wire in the disc-top counterbore	1 vial
Split pin / safety wire / zip-tie (option A only)	$\varnothing 1.0$ mm — passes through the standoff's tip cross-bore to lock the disc	6

Wire material trade-offs (pick one):

Material	$\varnothing$	Size vs λ at 10 MHz	Echo strength	Saturation risk on PV .035	Handling	Cost	When to choose
Nylon monofilament (4-0 polyamide suture, or 1-2 lb-test clear nylon fishing line)	70–100 μ m	0.5–0.7 λ	moderate (~ 6 dB single-surface reflection)	low — stays within the displayed dynamic range across a wide gain sweep	easy: pre-tensioned, doesn't sag, doesn't kink, can be heat-melted into a retaining ball at each disc face if you want a CA-free lock	low ($\sim$ \$5/spool)	Default choice for the PV .035 (10 MHz) — gives the cleanest PSF data because it doesn't saturate, doesn't ring, and is sub- λ axially
36 AWG enamel-coated copper magnet wire	127 μ m	$\sim 0.85 \lambda$	high	high — saturates by gain $\approx$ 50 in the existing P_035 captures	soft copper can yield under tension; permanent kinks if mishandled	low ($\sim$ \$5/spool)	If nylon is unavailable AND your captures will use low gain (≤ 40) AND axial PSF broadening from the $\sim \lambda$ -size diameter is acceptable
Tungsten wire	25–50 μ m	0.17–0.34 λ	very high (specular metallic reflection)	very high — saturates at any reasonable gain on the PV .035	brittle; awkward to thread; requires fine drill for a tight hole	medium (\$30–80/spool small min orders)	Only if you have a 20 MHz catheter (where 25 μ m = $\lambda/3$ fits the canonical AIUM spec); not recommended for the PV .035
Stainless steel / piano wire	50–100 μ m	0.34–0.68 λ	high	medium-high	stiff, holds tension well	low–medium	If nylon is unavailable and you need the wire to stay perfectly straight under load with no creep

Assembly:

1. Inspect both printed discs; clear each $\varnothing 0.5$ mm wire hole with a 0.5 mm drill bit (twist by hand, do not power-drill). The CA-cup counterbore should sit cleanly on the **top** face of each disc.
2. Stack the discs flat, hole-pattern aligned. Insert the 3 standoffs through the 3.2 mm BCD holes: - **Option A (printable standoffs)**: the $\varnothing 3.1$ mm tip pin slides through the disc; the $\varnothing 7$ mm middle shaft acts as a shoulder against the disc face. After both discs are seated, lock each pin with a split pin or zip-tie through the $\varnothing 1.5$ mm cross-bore at the pin tip. - **Option B (commercial M3 standoffs)**: screw both ends with M3 $\times$ 6 screws so the discs are parallel and 50 mm apart.

Verify with calipers that the disc-to-disc spacing is 50 ± 0.2 mm at all 3 standoff positions.

3. **Wire stringing.** Working from the outside ($n = 12$, $r = 26$ mm) to the inside ($n = 1$, $r = 4$ mm) so adjacent wires do not catch each other during threading: - Cut a 250 mm length of the chosen filament. (Magnet wire only: lightly sand the last 5 mm at one end to remove the enamel insulation if you plan to measure DC continuity for QC.) - Thread one end through the corresponding hole in the **bottom** disc and secure it taut to the underside (tape, mass, or a heat-melt ball for nylon). - Thread the other end through the matching hole in the **top** disc. - Pull by hand to remove sag — target tension ≈ 1 N. Tension can be judged by the wire's first-mode vibration frequency f_1 for a 50 mm span ($f_1 \approx 100$ Hz for 36 AWG copper, ≈ 70 Hz for 75 μ m nylon monofilament, ≈ 250 Hz for 25 μ m tungsten). - Lock the wire at the top face. Two options: * **CA lock (works for any material)**: hold tension and place a small bead of thin-CA into the top counterbore. The cup contains the bead and prevents it from running across the disc surface. After cure (≈ 30 s) lock the bottom exit the same way. * **Heat-melt ball (nylon only)**: hold tension and quickly touch a fine-tip soldering iron (≈ 250 °C) to the wire stub above the counterbore — the nylon balls up and forms a retaining knot that seats into the cup. Reversible (cut the ball off and re-thread). Repeat at the bottom face.

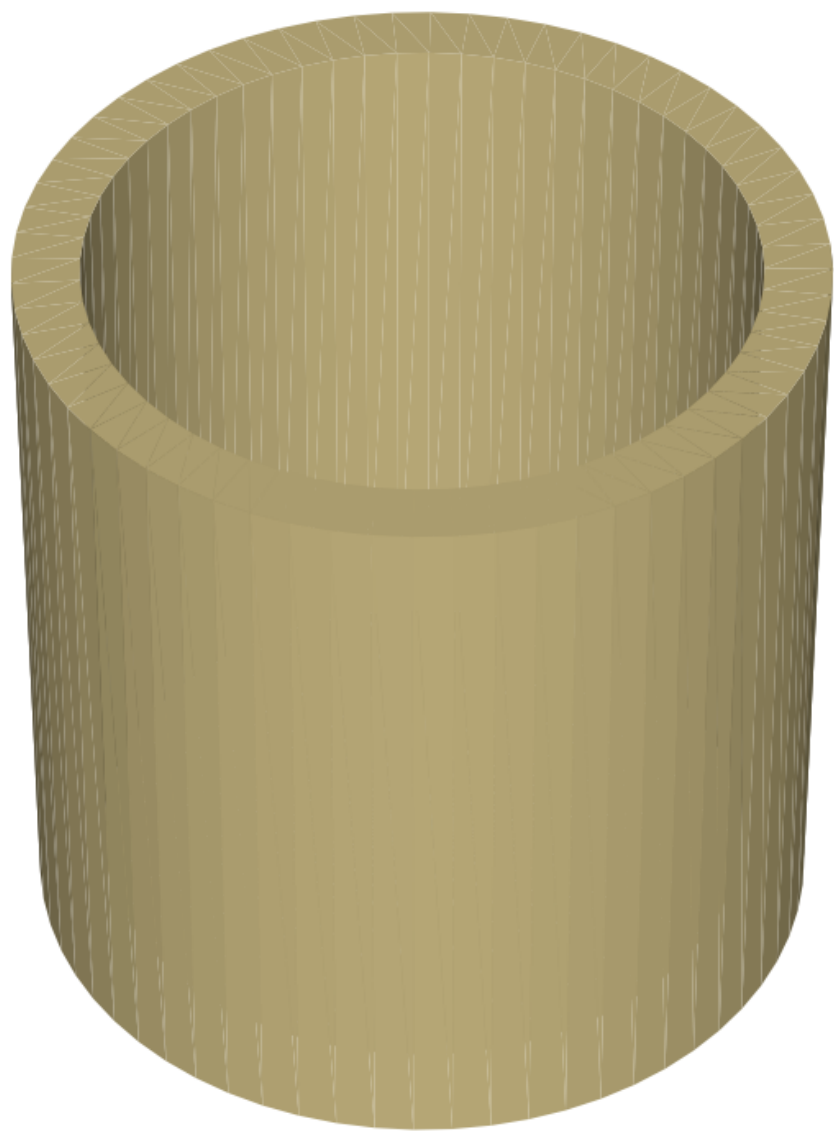
4. Repeat for all 12 wires.

5. **QC.** Photograph the assembly down the catheter axis with a USB microscope and 0.1 mm graticule: - All 12 wires must lie within ± 50 μ m of the design (r, θ) . - All wires must appear straight in both top-view and side-view photos (sag ≤ 50 μ m). - Document every wire's measured (r, θ) in `e2_wire_positions.csv`.

Catheter mounting: the catheter is fed up through the bottom disc's center hole, through the 50 mm wire span, and out the top disc. The disc holes are a slip-fit on the 1.9 mm OD Visions PV .035 catheter (0.05 mm radial clearance per side); the catheter is held coaxially by both end discs simultaneously. If your catheter measures > 1.95 mm OD with calipers, ream out the $\varnothing 2.0$ mm hole with a 2.0 mm twist drill and verify slip-fit before proceeding.

A.2 — Uniform tissue-mimicking phantom (E4)

Uniform-attenuation phantom mold (E4)



STL file: [hardware/phantom_mold_uniform.stl](#). Print 1 × (PETG, re-usable).

Recipe (Madsen-style agar / graphite, scaled to a 60 × 60 mm phantom):

Component	Mass / volume	Function	Tolerance
Distilled, degassed water	188 g	gel solvent	±1 g
Agar (high-strength, 600 g )	6 g (3 % w/w)	gel matrix	±0.05 g
n-Propanol (> 99 %)	14 g (7 % w/w)	sound-speed adjuster (target 1540 m/s)	±0.2 g
Graphite powder, 3 µm flake	4 g (2 % w/w)	acoustic scatterer + attenuator (target α = 0.5 dB/cm/MHz)	±0.05 g
Potassium sorbate	0.4 g (0.2 % w/w)	preservative	±0.05 g
Surfactant (Tween-20)	0.2 g (0.1 % w/w)	wet the graphite	drop count is fine

Theoretical α (recipe-only) ≈ 0.50 dB/cm/MHz, $c \approx 1540$ m/s, $\rho \approx 1030$ kg/m³, $Z \approx 1.59$ MRayl. This is sufficient for the **prior** in the parameter sheet but the actual α **must** be measured per A.6 if you want α to ± 0.10 dB/cm/MHz.

Equipment: 250 mL beaker, magnetic stirrer + hot plate (thermometer to ± 1 °C), digital scale (≥ 0.01 g resolution), vacuum chamber (≥ 70 kPa) or ultrasonic bath (40 kHz), printed mold A.2, $\varnothing 1.5$ mm $\times$ 80 mm PTFE rod (catheter channel former).

Procedure:

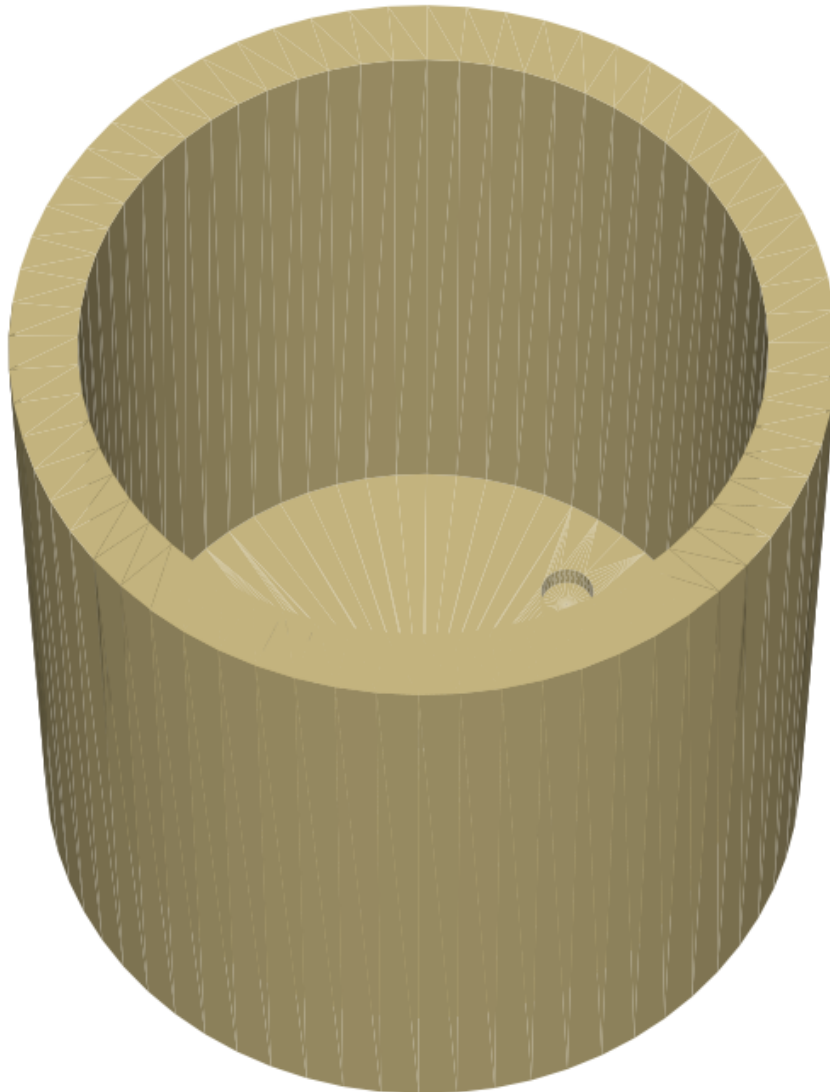
1. **Prepare the mold.** Lightly oil the mold interior with a release agent (silicone spray). Insert the 1.5 mm PTFE rod through the floor's $\varnothing 2.0$ mm bore, clamp it externally above and below the mold so it stands centered and vertical. Stand the mold on a level surface.
2. **Hydrate the agar.** Stir 6 g agar into 100 g of cold distilled water in the beaker; rest 5 min to fully wet.
3. **Disperse the graphite.** In a separate beaker, stir 4 g graphite + 0.2 g Tween-20 into 88 g of water until a uniform black slurry forms.
4. **Combine and heat.** Pour the graphite slurry into the agar slurry, add 0.4 g K-sorbate, stir continuously and bring the mixture to a gentle 90 °C boil on the hot plate (~10 min). Maintain 90 ± 2 °C for 5 min; the agar is now fully dissolved.
5. **Cool with stirring.** Remove from heat and stir while cooling to 60 °C (~5 min). Add the 14 g of n-propanol and stir an additional 1 min (do not boil after adding propanol).
6. **De-gas.** Pour into the vacuum chamber and pull ≥ 70 kPa for 5 min, or sonicate in the ultrasonic bath for 10 min, until visible micro- bubbles are gone.
7. **Pour the mold.** While still 50–55 °C, pour into the printed mold in one continuous pour. Avoid splashing onto the mold walls above the fill line. Tap the mold gently on the bench to release wall-bubbles.
8. **Set.** Cover with cling film and rest at 4 °C for ≥ 4 h. The phantom is ready when the surface is firm and clear.
9. **Demold:** trim with a scalpel along the lip; gently extract the PTFE rod by pulling vertically (the channel will be ~1.6 mm ID, slip fit on the 1.17 mm catheter).
10. **Storage.** Wrap in cling film + sealed bag at 4 °C; usable for 14 days. Add a 1 cm³ K-sorbate top-up gel layer if the surface dries.

Acceptance criteria:

- Visible homogeneity: no graphite settling layer at the bottom.
- Mass after demold within ± 2 % of the design mass (212.6 g) — confirms full liquid was retained.
- Channel concentricity ≤ 0.3 mm offset from the geometric center (measured by inserting a $\varnothing 1.5$ mm pin and visualizing through the gel).
- Independent α measurement (A.6) within ± 0.15 dB/cm/MHz of the recipe target.

A.3 — Cyst phantom (E5)

Cyst phantom mold (E5) Dimples in floor for 4 PTFE rods (4 mm OD)



STL file: [hardware/phantom_mold_cyst.stl](#). Same shell as A.2 but with 4 dimples in the floor at radii 10, 14, 14, 18 mm (azimuths 0°, 90°, 180°, 270°) for inserting cyst-forming PTFE rods.

Materials: A.2 gel recipe + four Ø4 mm × 60 mm PTFE rods. The 4 cyst rods press-fit into the floor dimples and stand vertically while the gel sets.

Procedure:

1. Prepare the mold as in A.2 (oil + insert Ø1.5 mm catheter rod).
2. Press the four Ø4 mm PTFE rods firmly into the floor dimples until bottomed; verify each is vertical with a small bubble level.
3. Cast the A.2 gel as in A.2 steps 2–8.
4. **Demold:** extract the catheter rod (vertical pull) and the four cyst rods (twist + vertical pull). The four cylindrical voids are the anechoic cysts.

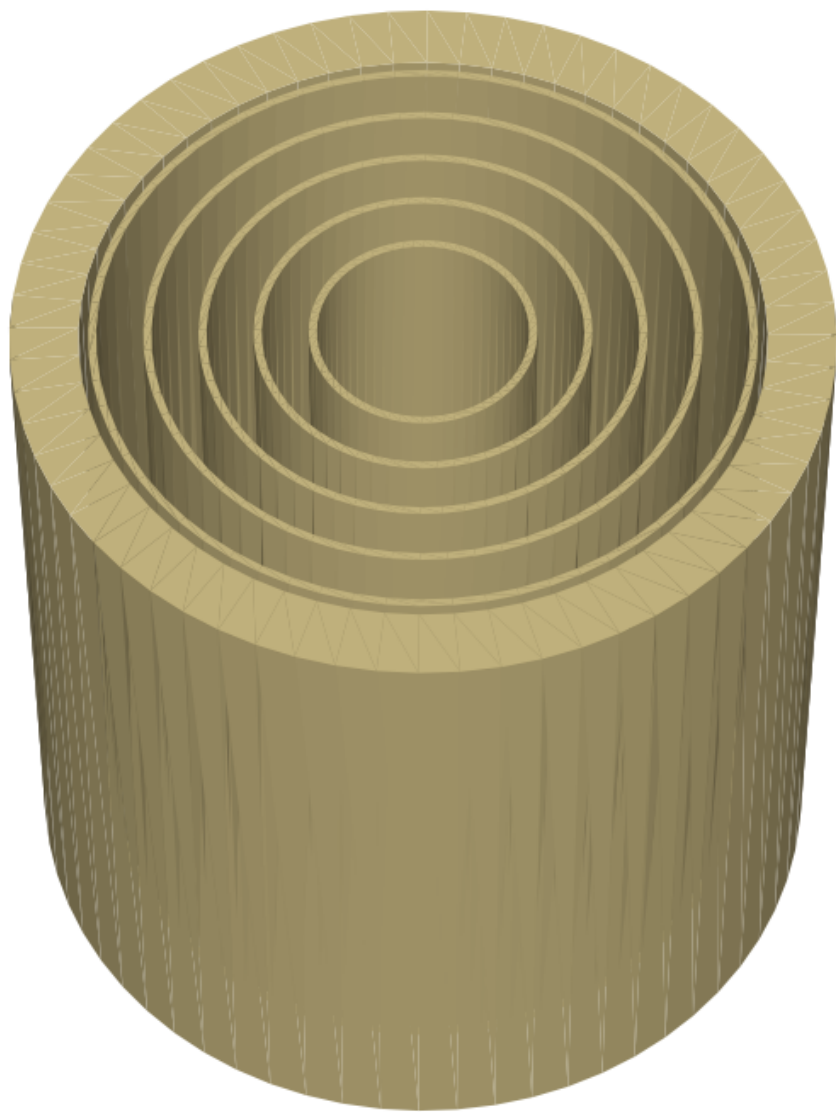
Acceptance criteria:

- All 4 cyst voids cleanly demolded, walls intact.
- Each cyst's diameter at the imaging plane within ± 0.2 mm of 4.0 mm (measured optically through the gel with a USB microscope + graticule card behind the phantom).
- Cyst-to-catheter radial distance (10, 14, 14, 18 mm) verified ≤ 0.3 mm error using the same optical setup.

If you want the standard CIRS 040GSE diameters (2 mm, 4 mm, 6 mm) instead of 4×4 mm, edit `CYST_ROD_DIA` and `CYST_RADII_MM` in `tools/gen_phantom_stl.py` and regenerate the STL. Use Ø2 mm and Ø6 mm PTFE rods accordingly.

A.4 — Step-attenuation phantom (E7 fallback)

Step-attenuation phantom mold (E7)
6 concentric chambers, 0.6 mm walls



STL file: [hardware/phantom_step_mold.stl](#). Single mold with **6 concentric chambers** separated by 0.6 mm integral walls at radii 8, 12, 16, 20, 24 mm. Each chamber is filled with a different graphite concentration to make a stepped-attenuation / stepped-grayscale target.

Recipe — vary graphite ONLY across chambers:

Chamber	Inner radii (mm)	Outer radii (mm)	Graphite (g/cc)	Target attenuation	Empirical vs. 0.1 mm
1 (center)	0.0	8.0	0.00	0.00	0.0
2	8.0	12.0	0.05	0.05	0.0
3	12.0	16.0	0.10	0.10	0.0
4 (center)	16.0	20.0	0.15	0.15	0.0
5	20.0	24.0	0.20	0.20	0.0
6 (outer)	24.0	28.0	0.25	0.25	0.0

All chambers use the same A.2 base recipe (water + agar + n-propanol + K-sorbate + Tween-20); only the graphite mass changes. Compute the new graphite mass for each chamber from the percentages above multiplied by the chamber's water mass:

Chamber	Approx. liquid volume (mL)	Water (g)	Graphite (g)
1	12.0	11.5	0.058
2	6.4	6.1	0.061
3	9.0	8.6	0.130
4	11.7	11.2	0.225
5	14.4	13.8	0.346
6	4.6	4.4	0.132

(Volumes assume 50 mm chamber height. Always batch $\geq 1.3\times$ the chamber volume to allow for spill / surface-tension losses.)

Procedure:

1. Print and prepare the mold; insert the catheter PTFE rod.
2. **Cast each chamber individually**, working from the **outer** chamber inward (chamber 6 $\rightarrow$ 5 $\rightarrow$ 4 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1). Pour each chamber, allow it to set partially (≈ 10 min at 4 °C — the chamber walls are very thin so the next pour cannot displace a partially-set neighbor) before pouring the adjacent inner chamber.
3. Use the smaller A.2 batch sizes (scale all reagents to the chamber's liquid volume).
4. After all chambers are poured, allow the assembly to set ≥ 4 h at 4 °C. **Do not demold**; the phantom is used in-mold (the printed walls are part of the acoustic structure but they are 0.6 mm thick PETG which contributes ≤ 0.5 dB extra attenuation between rings — document this in metadata if you need an absolute α calibration).

Acceptance criteria:

- Chamber-to-chamber boundary visible as a thin bright line in the IVUS image (the PETG wall) at the design radii (8, 12, 16, 20, 24 mm) within ± 0.3 mm.
- Mean grayscale of each chamber stratifies monotonically with graphite fraction.

A.5 — Layered tissue surrogate (E8)

E8 ideally uses real tissue (porcine LAD ex vivo, human cadaver coronary) to capture the per-material acoustic properties used by the simulator. A DIY surrogate cannot replace tissue for the final material database, but the recipes below are useful for **rehearsing the E8 procedure** and for sanity-checking the analysis pipeline before committing precious tissue samples.

Surrogate stack (cylindrical, 50 mm long):

Layer (radial extent)	Mimics	Recipe
0–3 mm	blood-filled lumen	water + 0.5 % graphite (very low scatter)
3–4 mm	intima	A.2 gel, 1.5 % graphite
4–6 mm	media	A.2 gel, 2.5 % graphite
6–7 mm	adventitia	A.2 gel, 3.5 % graphite + 5 % cellulose powder (high scatter)
7–10 mm	peri-adventitial fat	1:1 lard:gelatin (10 % w/w gelatin), no graphite

Procedure:

1. Print a 50 mm-tall, 25 mm-OD cylindrical mold (the A.2 mold can be re-used by inverting and machining out the floor; or print a custom open-ended tube — see [tools/gen_phantom_stl.py](#) to generate one).
2. Cast each layer in turn, **working radially outward**, using a series of removable PTFE collar tubes at each layer boundary (Ø6, 8, 12, 14 mm OD wall thickness ≤ 0.6 mm). After a layer sets, remove its bounding collar and pour the next layer directly against the already-set inner gel.
3. The lumen layer (0–3 mm) is poured *last* using a thinner gel that stays liquid long enough to fully wet the catheter channel.

This is a substantial multi-day cast; only attempt it if real tissue is unavailable.

A.6 — Independent α calibration (transmission-substitution)

For all DIY phantoms above, the recipe gives an α prior good to about ± 0.2 dB/cm/MHz. To tighten this you need a **one-time** independent measurement using two unfocused 20 MHz transducers in a transmission geometry. This calibration is performed **once per phantom batch**, not per experiment.

Equipment: two unfocused 20 MHz transducers (e.g. Olympus V317-SU, 6.35 mm dia.), broadband pulser/receiver (Olympus 5072PR or equivalent), 12-bit DAQ ≥ 100 MS/s, water tank with the two transducers facing each other 30 mm apart, one 10 mm-thick sample of the phantom material poured into a thin acrylic ring (cast separately at the same time as the main phantom), thermometer.

Procedure:

1. Place the two transducers face-to-face, 30 mm apart, aligned $\leq 0.1^\circ$ off-axis. Pulse from one, receive on the other; record the through- water reference RF $r_{\text{water}}(t)$.
2. Insert the 10 mm phantom-material slab between the transducers, reposition transducers so the through-distance is still 30 mm (sample slab + 20 mm water). Record $r_{\text{sample}}(t)$.
3. Fourier-transform both signals. Compute the per-frequency transmission coefficient $T(f) = |R_{\text{sample}}(f)| / |R_{\text{water}}(f)|$ and the per-frequency attenuation $\alpha(f) = -20 \cdot \log_{10}(T(f) \cdot 4 \cdot Z_w \cdot Z_s / (Z_w + Z_s)^2) / d_{\text{sample}}$ (the parenthesized term corrects for the two water-sample interface reflections; for graphite-loaded gel, $Z_s \approx 1.59$ MRayl, so the interface correction is ≈ 0.05 dB and can usually be neglected).
4. Fit $\alpha(f) = \alpha_0 \cdot f$ over 10–25 MHz; the slope is α in dB/cm/MHz.

Result: the calibrated α value goes into the [Source](#) field of the parameter table for E4 (replacing the recipe-prior value).

A.7 — Mold print settings & acceptance

Parameter	Setting
Filament	PETG (preferred) or ABS
Layer height	0.20 mm
Walls / perimeters	4
Top / bottom solid layers	6
Infill	30 % gyroid
Print speed	≤ 60 mm/s for outer perimeters
Bed temp	80 °C (PETG) / 100 °C (ABS)
First-layer flow	110 % (for water-tight floor)

Print acceptance:

- Floor is water-tight: fill with water for 10 min and confirm zero weeping at the floor seam.
- All design dimensions within ± 0.2 mm of nominal (calliper check on the outer diameter, inner diameter, and floor thickness).
- Catheter bore is straight (insert a $\varnothing 1.5$ mm steel pin; pin must drop through under its own weight).
- For the step mold (A.4), each integral divider wall is fully fused to the floor (visual check; no light gap when held to a lamp).